

Q1: Genetic Algorithm

A university is scheduling exams across 6 slots:

S1, S2, S3, S4, S5, S6

Penalty matrix (conflicts between exams):

	S1	S2	S3	S4	S5	S6
S1	0	4	7	3	6	5
S2	4	0	2	8	3	7
S3	7	2	0	6	4	3
S4	3	8	6	0	5	2
S5	6	3	4	5	0	1
S6	5	7	3	2	1	0

Parents:

- P1: [S2, S4, S1, S6, S3, S5]
- P2: [S5, S1, S3, S2, S6, S4]

a. Uniform Crossover

Pattern: ABBABA

Generate two children and fix duplicates.

b. Mutation

Apply swap mutation (positions 1 and 6)

c. Fitness

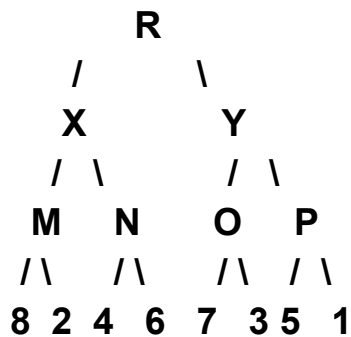
Calculate total penalty.

d. Improvement

Did mutation improve fitness?

Q2: Alpha Beta Pruning

Tree:



Tasks:

a. Compute Minimax value

b. Apply Alpha-Beta Pruning (left → right)

Mark pruned branches

c. Write order of node evaluation